Statistics Lab (52568)

Semester Summary: Genomic Coverage, GC Bias, and Copy-Number Estimation

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Contents

1 The Big Picture: One Problem, Seven Labs

1.1 Biological motivation

Our genetic information is encoded in DNA molecules arranged as chromosomes. In a healthy human cell there are 46 chromosomes: 22 homologous pairs plus one sex-chromosome pair (XX or XY). The double-helix consists of two parallel strands of nucleotides (bases A, C, G, T) held together by complementary pairing: A↔T and C↔G. When a cell divides, the double helix unzips and each strand is used as a template to synthesize its complement, so each daughter cell receives an identical copy.

Copy number variation (CNV) refers to regions where this copying process goes wrong. A segment can be duplicated (*copy-number gain*) or deleted (*copy-number loss*), changing the number of copies of that genomic region from the normal 2. CNVs are part of normal human variation, but they are far more frequent in cancer, where the machinery controlling cell division is damaged:

- **Down syndrome** (trisomy 21) is caused by a third copy of chromosome 21, disrupting developmental balance.
- In **tumors**, copy-number gains of growth-promoting genes can drive uncontrolled proliferation. A 2005 study found that a growth hormone gene was amplified in the majority of nine tumor biopsies examined.

Detecting CNVs is therefore a clinically important problem. We measure them indirectly through **sequencing coverage**.

1.2 High-throughput DNA sequencing

Next-generation sequencing (NGS, e.g. Illumina/Solexa) allows millions of short DNA fragments to be read in a single run. The pipeline has three stages:

1. **Library preparation**: genomic DNA is fragmented into pieces of roughly uniform size and *amplified by PCR* (copies made so there is enough material to sequence).
2. **Sequencing**: each fragment is immobilized, copied base-by-base with fluorescently labelled nucleotides, and the color signal is read — yielding a short *read* of ∼70–100 bases from each end. The output is a FASTQ file with read sequences and quality scores.
3. **Alignment (mapping)**: reads are aligned back to a reference genome (we use `hg18`, a consensus built by the Human Genome Project, 2000) using fast hash-based algorithms. The output is a SAM/BAM file storing the chromosomal location and length of each mapped fragment.

We work with a filtered version of this output: a `.forward` file containing `(Chromosome, Location, Length)` for one paired-read direction on chromosome 1. Our patient is `TCGA-13-0723`, with a healthy tissue sample (`-10B`) and a paired tumor sample (`-01A`).

Coverage as a proxy for copy number. The number of reads mapping to a region is proportional to the number of DNA copies of that region in the sample: if a segment is duplicated (4 copies instead of 2), it should on average produce twice as many reads. For a single sample:

$$\frac{\text{reads at region}}{\text{reads at typical region}} \approx \frac{\text{copy number}}{2}.$$

For two samples (tumor vs. normal), the ratio of read counts at a locus is proportional to their copy-number ratio, assuming otherwise uniform coverage. The central difficulty — and the theme of the entire course — is that coverage is *not* uniform: it is systematically distorted by the local GC content of the DNA sequence, a bias introduced largely by the PCR step.

1.3 The semester arc

Lab	Topic	What it adds
1	Base composition	Describe the raw sequence; see Chargaff pairing; find high-GC region
2	Coverage computation	Efficient read-counting; chromosome-wide coverage map; distribution
3	Poisson assessment	Test the uniform model; discover overdispersion; find GC as the cause
4	Spline regression	Model $f(\text{gc})$; residual analysis; learning curves
5	GC analysis (AI-first)	G vs. C vs. G+C; spatial consistency; bin size; Poisson GLM
6	Copy-number estimation	Plug-in estimator; variance decomposition; noise models
7	Tumor vs. healthy	Two-sample comparison; aggregative bin-size selection
8	GC correction review	(covered within Lab 9 analysis)
9	Anomaly detection	Hypothesis testing; empirical null; FDR; CNV segmentation

2 Part I — What We Learned

2.1 From reads to coverage

The first computational lesson was that *correctness and speed are separate concerns*: always start from a slow version you trust, then optimize. Counting reads per base can be done naively (loop over every base, scan for matching reads) or efficiently with R's vectorized `tabulate()` function, which exploits the fact that read locations are already sorted. Figure 1 shows the performance difference over 100 random 1-Mb windows.

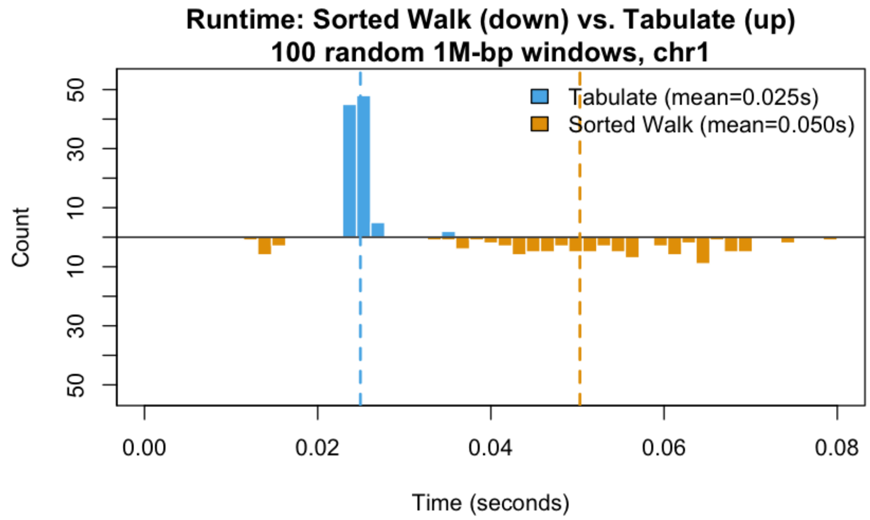

Figure 1: Lab 2 — Algorithm benchmark. Back-to-back histogram of runtimes over 100 random 1-Mb windows. `tabulate` (blue, up) averages 0.025 s; sorted walk (orange, down) averages 0.054 s — more than twice as slow.

Binning. Base-level coverage is extremely sparse (most bases have 0 or 1 read). We aggregate reads into *bins* of width b, so $Y_k = \sum_{\text{bases in bin}} Y_i$ is the working unit for all subsequent modelling. Bin size is a genuine modelling choice with real tradeoffs (§2.7).

2.2 The uniform Poisson model and its assumptions

The simplest model imagines reads landing like raindrops: each of the J fragments independently picks a start base, uniformly at random over the I bases. This rests on three explicit assumptions:

1. **Identical distribution** — every fragment has the same landing probabilities.
2. **Uniformity** — $p_i = 1/I$ for all bases i.
3. **Independence** — fragments land independently.

Under (1)+(3), $Y_i \sim \text{Binomial}(J, p_i)$. Adding (2) and the Poisson approximation (large J, tiny $1/I$):

$$Y_i \sim \text{Poisson}(\lambda), \qquad \lambda = J/I.$$

Because independent Poissons add, a bin satisfies $Y_k \sim \text{Poisson}(\lambda_k)$. The defining testable property:

$$\mathbb{E}[Y] = \text{Var}[Y] = \lambda \quad \Longrightarrow \quad \text{VMR} = \frac{\text{Var}[Y]}{\mathbb{E}[Y]} = 1.$$

SD grows like $\sqrt{\lambda}$, so relative noise shrinks as bins grow — a first taste of the bias–variance tradeoff.

2.3 Why the uniform model fails: overdispersion and the variance decomposition

Real coverage is heavily **overdispersed**: VMR ≈ 14.7 rather than 1. The theoretical tool that explains *why* is the **law of total variance**, applied to the natural two-level structure of our data.

The two-level model. We are *not* claiming all bins share a single rate λ. The more honest description is that each bin k has its own rate:

$$Y_k \sim \text{Poisson}(\lambda_k),$$

where λ_k varies across the genome (driven by local sequence properties we have not yet modeled). Each bin still satisfies the Poisson property *conditional on its own rate*:

$$\mathbb{E}[Y_k \mid \lambda_k] = \lambda_k, \qquad \text{Var}(Y_k \mid \lambda_k) = \lambda_k.$$

The question is: what is the *marginal* variance of Y_k when we also account for the variation of λ_k across bins?

The law of total variance. For any two random variables Y and X:

$$\text{Var}(Y) = \underbrace{\mathbb{E}\big[\text{Var}(Y \mid X)\big]}_{\text{average within-group variance}} + \underbrace{\text{Var}\big(\mathbb{E}[Y \mid X]\big)}_{\text{variance of group means}}.$$

This is an exact identity — no approximation or model assumption beyond the existence of these moments. Setting $Y = Y_k$ and $X = \lambda_k$:

$$\text{Var}(Y_k) = \mathbb{E}\big[\text{Var}(Y_k \mid \lambda_k)\big] + \text{Var}\big(\mathbb{E}[Y_k \mid \lambda_k]\big).$$

Substituting the two Poisson identities:

$$\boxed{\text{Var}(Y_k) = \underbrace{\mathbb{E}[\lambda_k]}_{\text{Poisson term (within-bin noise)}} + \underbrace{\text{Var}(\lambda_k)}_{\text{rate variation (between-bin heterogeneity)}}}$$

Writing this empirically over the K observed bins (replacing population moments with sample averages):

$$\text{Var}(Y_k) \approx \bar{\lambda} + \frac{1}{K}\sum_{k=1}^{K}(\lambda_k - \bar{\lambda})^2, \qquad \bar{\lambda} = \tfrac{1}{K}\textstyle\sum_k \lambda_k.$$

Why VMR $= 1$ characterizes the uniform model. Dividing both sides by the mean:

$$\text{VMR} = \frac{\text{Var}(Y_k)}{\mathbb{E}[Y_k]} = 1 + \frac{\text{Var}(\lambda_k)}{\bar{\lambda}}.$$

The VMR equals 1 **if and only if** all rates λ_k are identical — which is exactly the uniform Poisson assumption. Any spatial heterogeneity in rates pushes VMR above 1. Our observed VMR ≈ 14.7 means:

$$\text{Var}(\lambda_k) \approx 13.7 \cdot \bar{\lambda},$$

i.e. the rate-variation term is roughly $14\times$ larger than the Poisson noise.

Why the rate-variation term is so large. One might expect that summing 5000 bases into a bin would average out rate fluctuations (a central-limit argument). But this only works if neighboring bases have *independent* rates. In reality, nearby bases have similar GC content, so their λ values are positively correlated. When correlated quantities are summed, variance does not shrink — the rate variation persists at the bin scale. This is exactly what GC content creates: spatially structured, persistent rate heterogeneity that does not wash out under aggregation.

Summary of the terms.

Term	Interpretation	Vanishes when...
$\bar{\lambda}$	Irreducible Poisson sampling noise	Never ($\lambda > 0$)
$\mathrm{Var}(\lambda_k)$	Rate heterogeneity across the genome	All λ_k are equal (uniform model)

This decomposition is the conceptual engine of the semester: the course is about *modeling the second term* (via GC content) and removing it, so that coverage reflects copy number (a biological signal) rather than sequencing bias.

Figure 2 shows the observed vs. Poisson distribution (Lab 3), and Figure 3 shows the GC–coverage scatter that explains the discrepancy.

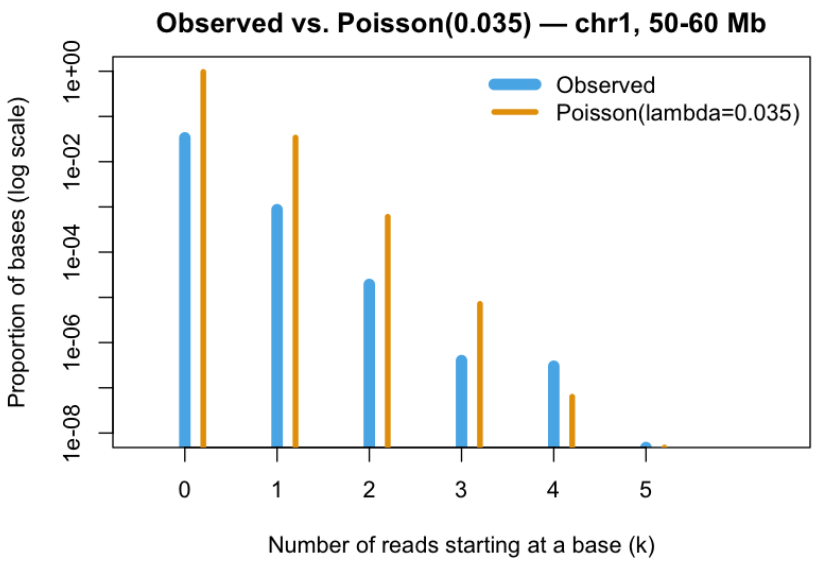

Figure 2: Lab 3 — Observed vs. Poisson distribution. Blue: observed proportions of bases with k reads (log scale). Orange: Poisson($\lambda = 0.035$) prediction. The observed distribution is clearly less left-skewed and more right-skewed than Poisson, confirming the model fails.

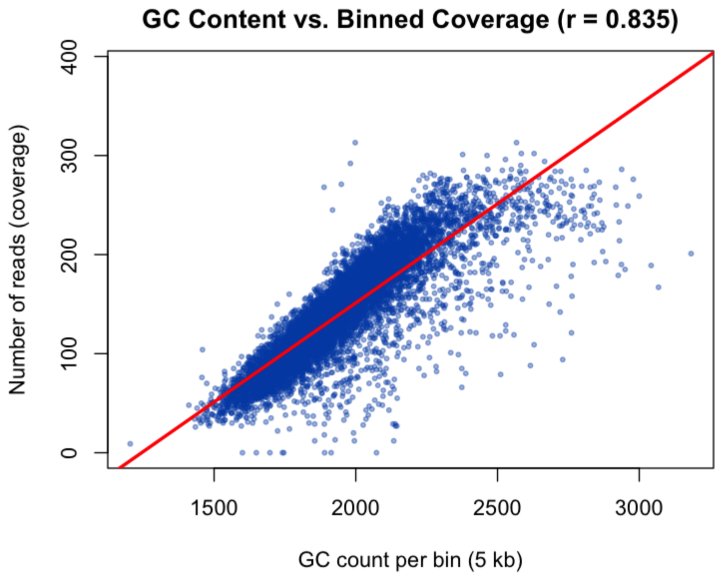

Figure 3: Lab 3 — GC content vs. binned coverage (50–100 Mb, 5 kb bins). Strong positive linear trend ($r \approx 0.84$, red line). GC-rich bins attract systematically more reads, violating the uniformity assumption.

2.4 The GC-content bias and the Benjamini & Speed (2012) paper

This is the central phenomenon of the course, and of our professor's own paper: **Benjamini & Speed (2012), "Summarizing and correcting the GC content bias in high-throughput sequencing,"** ***Nucleic Acids Research*** **40(10):e72.**

Key findings from the paper:

- The relationship is **unimodal**: both GC-rich *and* AT-rich fragments are underrepresented; coverage peaks at intermediate GC (≈50%). (At our 50–100 Mb binned scale we see mostly the rising left arm, but the spline captures the curvature.)
- The GC content of the **entire fragment** (not just the sequenced read) drives the bias most strongly — evidence that **PCR amplification**, not sequencing chemistry, is the main cause.
- After correcting for GC, copy-number gains hidden by the bias become visible. The paper demonstrates recovery of a CN gain on chromosome 2 in tumor data.

The course implements this correction step by step, using the same conceptual framework: model coverage as $a_k \cdot f(\mathrm{gc}_k)$, estimate f, divide it out.

2.5 The full-chromosome view

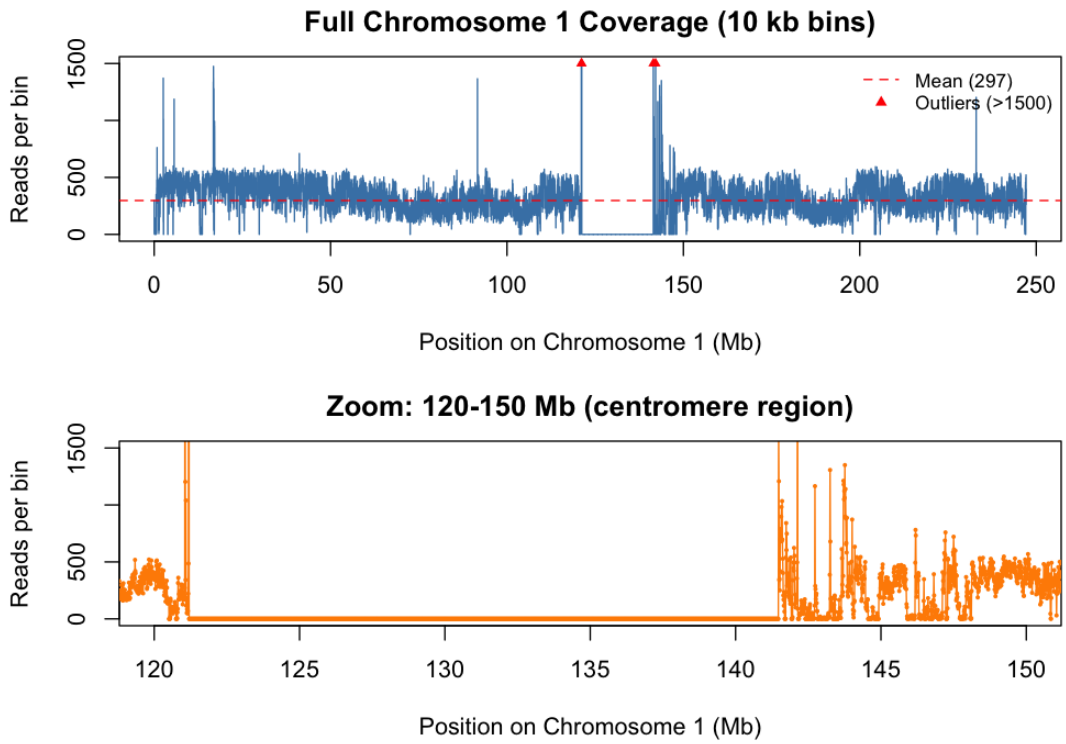

Figure 4: Lab 2 — Full chromosome coverage and centromere zoom. *Top:* Coverage across all of chromosome 1 (10 kb bins). The dashed red line is the mean. Black vertical lines mark the zoom region. Red triangles mark outlier bins clipped at 1500 reads. Note the coverage gap at ∼120–130 Mb (centromere) and extreme spikes at ∼130–145 Mb (likely duplicated segments). *Bottom:* Zoom-in on 120–150 Mb showing the centromere gap clearly.

2.6 Regression: linear, then non-parametric (splines)

We first modelled f linearly via `lm`: $Y_k = \beta_0 + \beta_1\, \mathrm{gc}_k + \epsilon_k$. This introduced the regression toolkit: coefficients and standard errors, residual plots, RMSE, the danger of outliers, and the formula $\hat{\beta}_1 = \widehat{\mathrm{Cov}}(\mathrm{gc}, Y)/\widehat{\mathrm{Var}}(\mathrm{gc})$.

The true relationship is curved (Figure 5), motivating **spline regression**. The key ideas, built up from first principles in lecture:

- Partition the predictor range at **knots**; fit a polynomial in each piece.
- Enforce smoothness at knots: a **cubic regression spline** keeps the function and its first two derivatives continuous, using truncated-power basis functions $f_d(x, t) = (x - t)^d \cdot \mathbf{1}\{x > t\}$.
- B-splines (`bs()` in R) give the same fit with a better-conditioned, locally supported basis.
- Modelling choices: **knot count/placement** and **degree**. More knots ⇒ more flexibility; higher degree ⇒ smoother curves.
- Despite the non-parametric spirit, the model is fit by ordinary OLS on the basis columns — so all standard regression tools (residuals, R^2, AIC) apply.

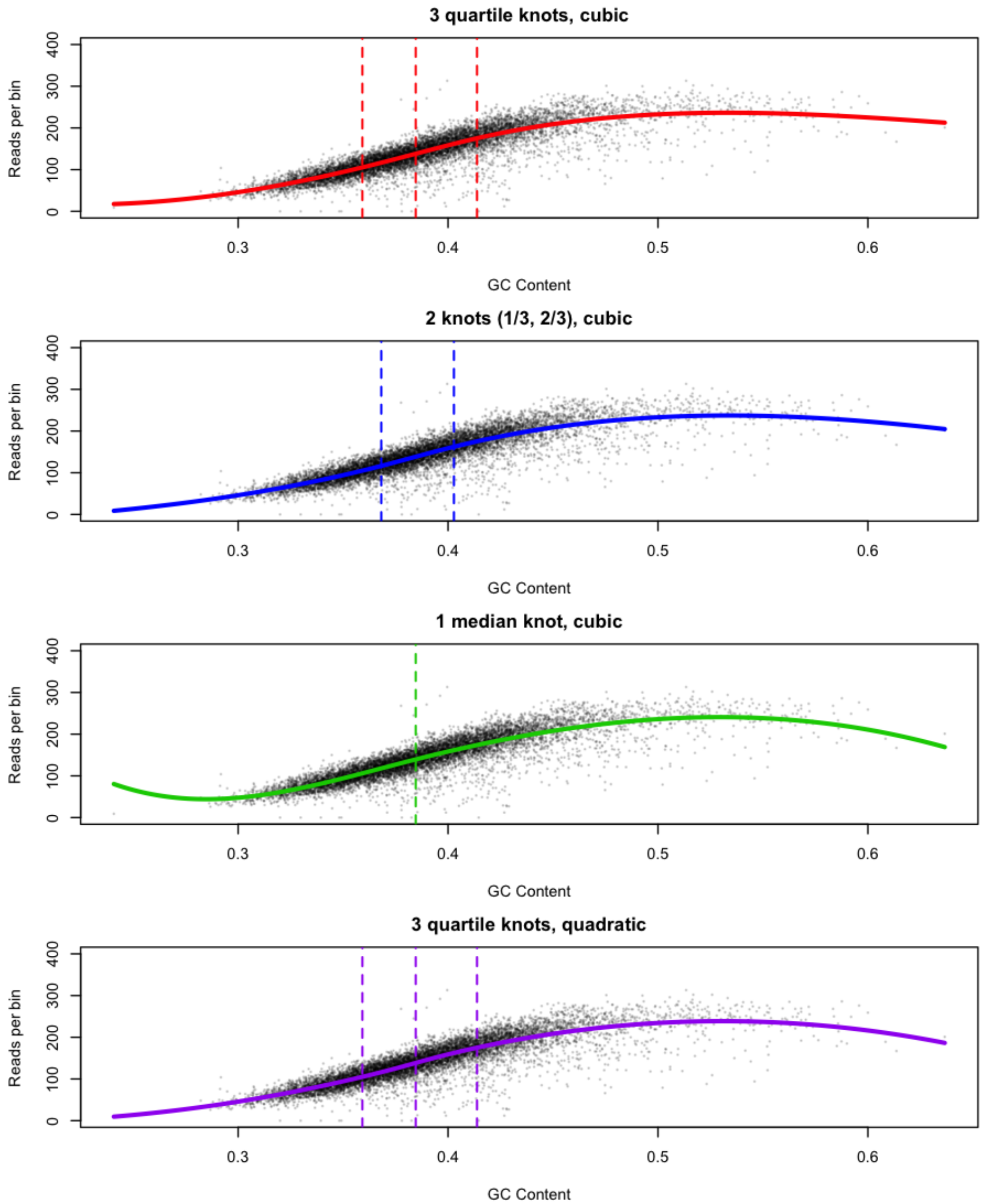

Figure 5: Lab 4 — Four B-spline models for GC–coverage. Each panel shows GC content (x) vs. reads per 5 kb bin (y, gray points) with the fitted spline (colored line) and knot positions (dashed verticals). All four models capture the main trend similarly; the models diverge in the high-GC tail where data are sparse. $R^2 \approx 0.75$ for all candidates.

2.7 Model selection, generalization, and the bias–variance tradeoff

Flexible models risk overfitting, so we judge them **out-of-sample**: train/validation/test splits (`set.seed` for reproducibility), **learning curves** (validation RMSE vs. training-set size), and robust metrics alongside RMSE (median, IQR, median absolute error).

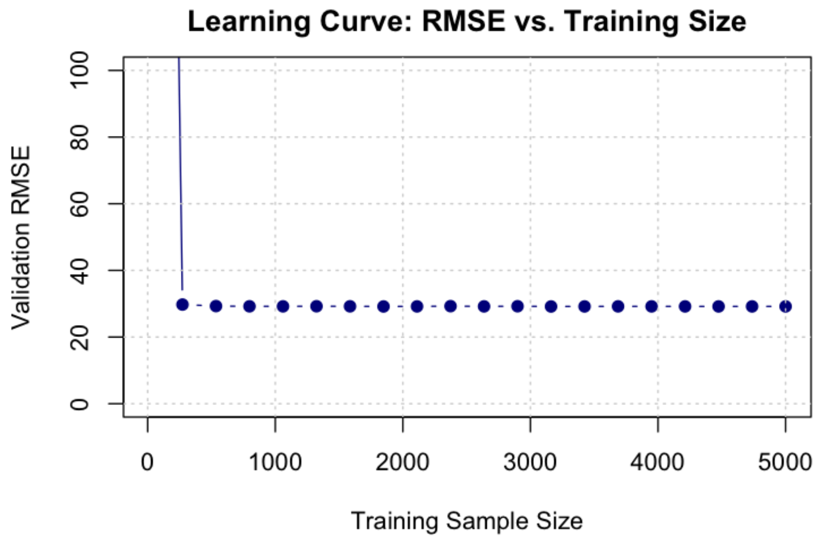

Figure 6: Lab 4 — Learning curve. Validation RMSE as a function of training sample size. RMSE drops sharply from ∼100 at $n = 10$ and plateaus near 30 by ∼500 samples, showing the GC trend is learnable with little data.

The **bias–variance tradeoff** appears most clearly in **bin size**. Larger bins average out per-base Poisson noise (lower variance) but blur the local GC signal and smear genomic features (higher bias). Figure 7 shows the marginal R^2 gain from increasing bin size. The statistical *elbow* falls near 5 kb; combined with the scale of biological features (genes 10–100 kb, CpG islands 0.5–2 kb), 5 kb is a principled choice.

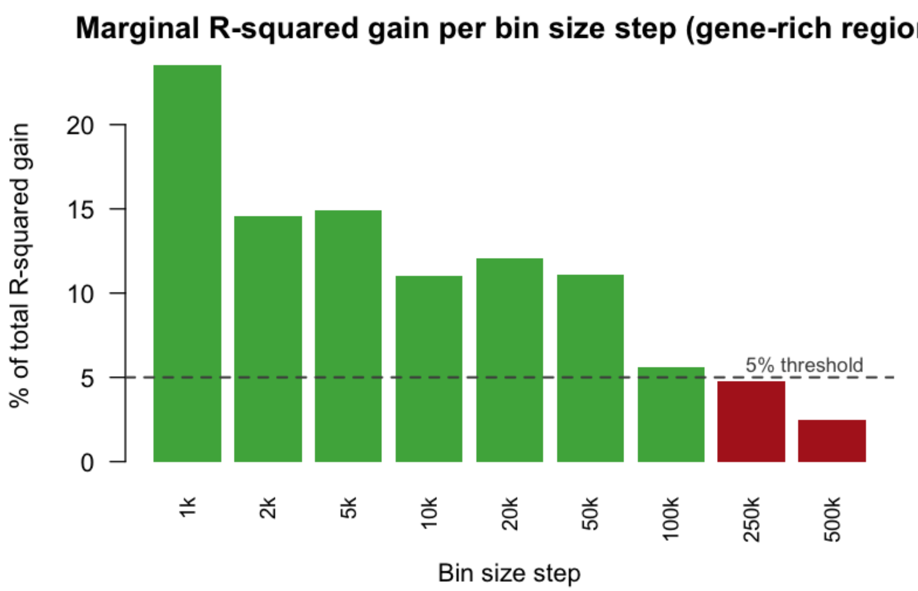

Figure 7: Lab 5 — Marginal R^2 gain per bin-size step (gene-rich region). Green bars contribute ≥5% of the total gain; red bars represent diminishing returns below the threshold. The elbow falls at ∼5 kb: beyond this, more aggregation costs spatial resolution without meaningfully improving the model.

2.8 GC-skew and why G+C beats either alone

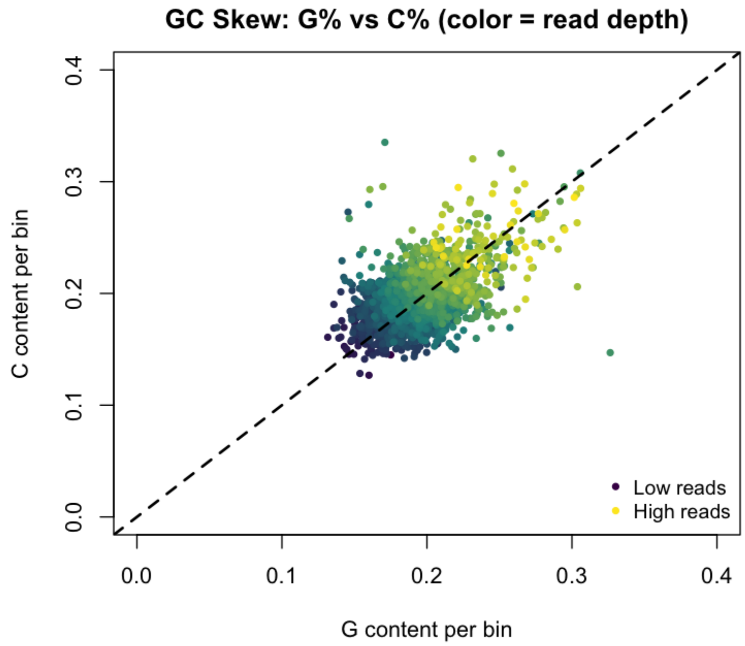

Figure 8: Lab 5a — GC skew: G% vs C% per bin, color = read depth. The cloud follows the Chargaff diagonal (G% ≈ C% globally), but fans out at higher GC values. High-GC bins (yellow = high reads) spread away from the diagonal due to transcription-coupled strand asymmetry in gene-rich regions. This means C carries independent signal precisely where coverage is highest — hence `~G+C` outperforms `~G` alone by ≈0.17 in R^2.

2.9 Spatial consistency of the GC–coverage relationship

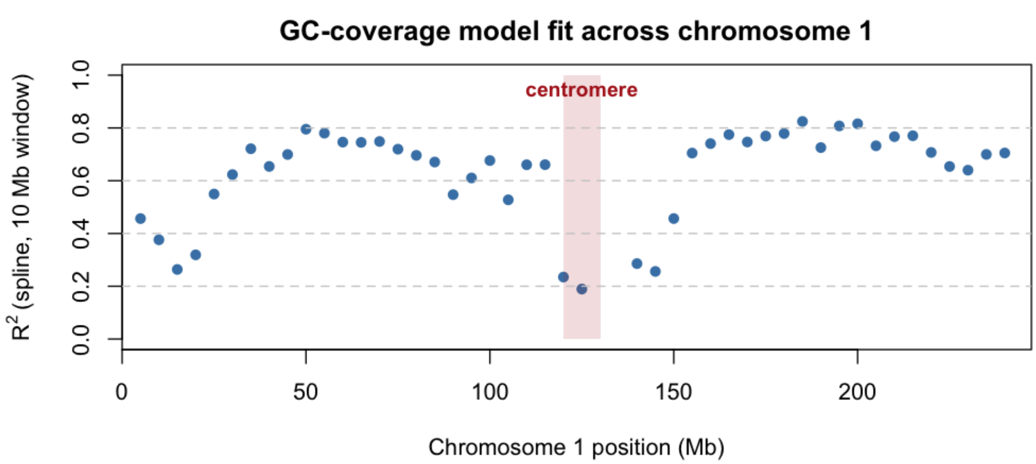

Figure 9: Lab 5b — Spatial R^2 profile across chromosome 1. Each dot is the spline model R^2 for a 10 Mb window. Gene-rich regions reach $R^2 \approx 0.7$; the centromere (∼120–130 Mb, shaded red) drops to ≈ 0.3. The relationship is not spatially uniform — per-region fitting is justified.

2.10 Poisson GLM and quasi-Poisson

Because the response is a count, the natural regression is a **generalized linear model** with log link: $\log \mathbb{E}[Y_k] = \beta_0 + \beta_1 \, \mathrm{gc}_k$. Plain Poisson GLM forces $\mathrm{Var}(Y) = \mathbb{E}[Y]$, which we know is violated. **Quasi-Poisson** relaxes this to $\mathrm{Var}(Y) = \phi \, \mathbb{E}[Y]$ and estimates dispersion ϕ directly. The two give nearly identical fitted means but different uncertainty estimates. The residual $\phi \approx 14$ after including GC confirms substantial overdispersion remains — GC explains part but not all of the extra variance.

2.11 The copy-number estimation model

Everything converges here. The generative model is:

$$Y_k \sim \mathrm{Poisson}(\lambda_k), \qquad \lambda_k = a_k \cdot f(\mathrm{gc}_k) + \epsilon_k,$$

with identifiability constraints: $\mathrm{med}_k\{a_k\} = 1$, $\mathbb{E}_k[f(\mathrm{gc}_k)] = 1$, $\mathbb{E}[\epsilon_k] = 0$.

Solving for a_k and substituting our spline estimate $\hat{f}$ gives the **plug-in estimator**:

$$\hat{a}_k = \frac{y_k}{\hat{f}(\mathrm{gc}_k)} \cdot \hat{M}, \qquad \hat{M} = \frac{1}{\mathrm{med}_k \left\{ \dfrac{y_k}{\hat{f}(\mathrm{gc}_k)} \right\}},$$

i.e. *divide out the GC effect, then renormalize* so the median estimate is exactly 1. Normal genomes should concentrate around 1; departures flag CN events.

2.12 Variance decomposition of the estimator

To leading order, $\mathrm{Var}(\hat{a}_k) \approx \left[\lambda_k + \sigma^2\right]/f(\mathrm{gc}_k)^2$, where the first term is irreducible Poisson sampling noise and the second is technical ϵ noise (σ^2 estimated from spline residuals, no normality assumed). Their relative shares tell us whether **more reads** (shrinks the Poisson term) or a **better technical model** (shrinks the ϵ term) is the path to improvement.

2.13 The two-sample noise decomposition (Yuval's theory note)

Yuval's second copy-number note provides a richer two-sample model. The noise in a single sample is decomposed into a **shared component** γ_k (same in both tumor and healthy, e.g. local mappability, chromatin structure) and **sample-specific** components δ_k^T, δ_k^N:

$$Y_k^T = a_k \cdot f^T(\mathrm{gc}_k) \cdot \gamma_k \cdot \delta_k^T + \eta_k^T, \qquad Y_k^N = f^N(\mathrm{gc}_k) \cdot \gamma_k \cdot \delta_k^N + \eta_k^N.$$

The two-sample estimator

$$\hat{a}_k^{(2)} = \frac{Y_k^T / \hat{f}^T(\mathrm{gc}_k) + c}{Y_k^N / \hat{f}^N(\mathrm{gc}_k) + c} \approx a_k \cdot \frac{\delta_k^T}{\delta_k^N}$$

cancels the shared component γ_k entirely — what remains is only the ratio of the sample-specific noises δ_k^T/δ_k^N. This is the key advantage over single-sample correction: if the dominant technical variation is shared (mappability, GC curves of the same shape), the two-sample ratio is much cleaner than either sample alone.

The stabilized single-sample estimator $\hat{a}_k^{(1*)} = (Y_k + c)/(\hat{f} + c)$ trades a small increase in bias (for bins where $a_k \neq 1$) for reduced variance at low-coverage bins.

2.14 Additive vs. multiplicative noise

Should the noise be **additive** ($\lambda_k = a_k f + \epsilon$, fixed-size offset) or **multiplicative** ($\lambda_k = a_k f \cdot \eta$, $\eta \sim N(1, \gamma^2)$, noise scaling with signal)? Biologically, the multiplicative form is more plausible: PCR amplifies errors in proportion to the amount of DNA, so larger expected coverage carries larger absolute fluctuations — consistent with the heteroscedasticity we observed (residual spread grows with GC/coverage).

2.15 Two-sample comparison: tumor vs. healthy

With a healthy sample as baseline (copy number ≈ 1 everywhere) and a paired tumor sample, the GC correction is fit *separately per sample* (the bias function differs between libraries). After per-sample correction and median normalization, bins where tumor and healthy CN diverge are candidate CN events. This is exactly the paper's demonstration: a copy-number gain hidden by GC bias becomes visible after correction.

3 Part II — Our Lab Results

3.1 Lab 1 — Base composition of chromosome 1

We counted A/C/G/T in 1000-bp cells over 50–70 Mb. Running medians ($k = 201$, Figure 10) confirmed **Chargaff pairing**: A and T track each other, as do C and G. Pair scatter plots (Figure 11) showed the strongest positive correlations for A–T and C–G. We flagged a **high-GC anomaly at cells ∼3000–5500 (≈53–55.5 Mb)**, visible as the shaded region in the lab.

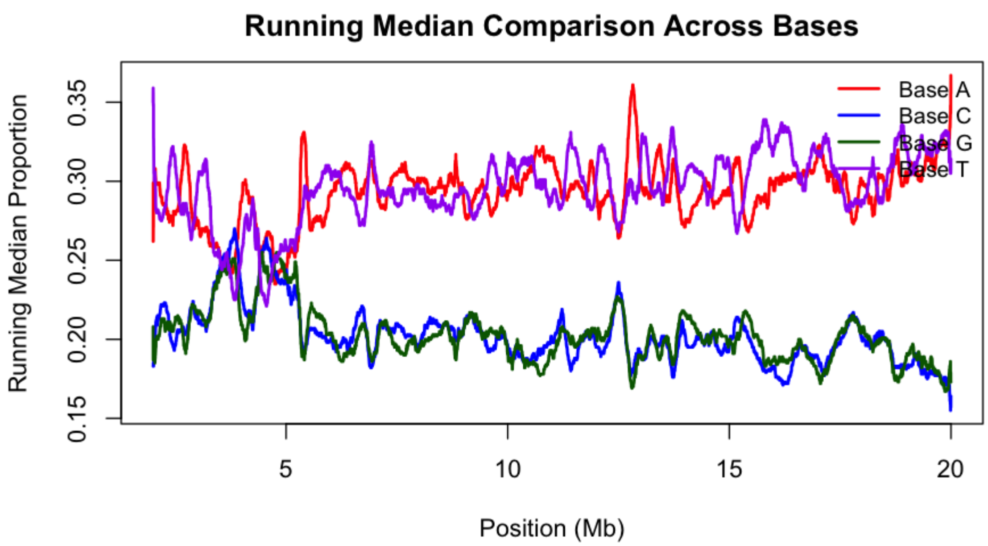

Figure 10: Lab 1 — Running median comparison across bases. Smoothed proportions of A (red), C (blue), G (dark green), and T (purple) along the chromosome. A≈T and C≈G throughout, confirming Chargaff's rule. C and G are systematically lower, consistent with a slight AT bias in the analyzed region.

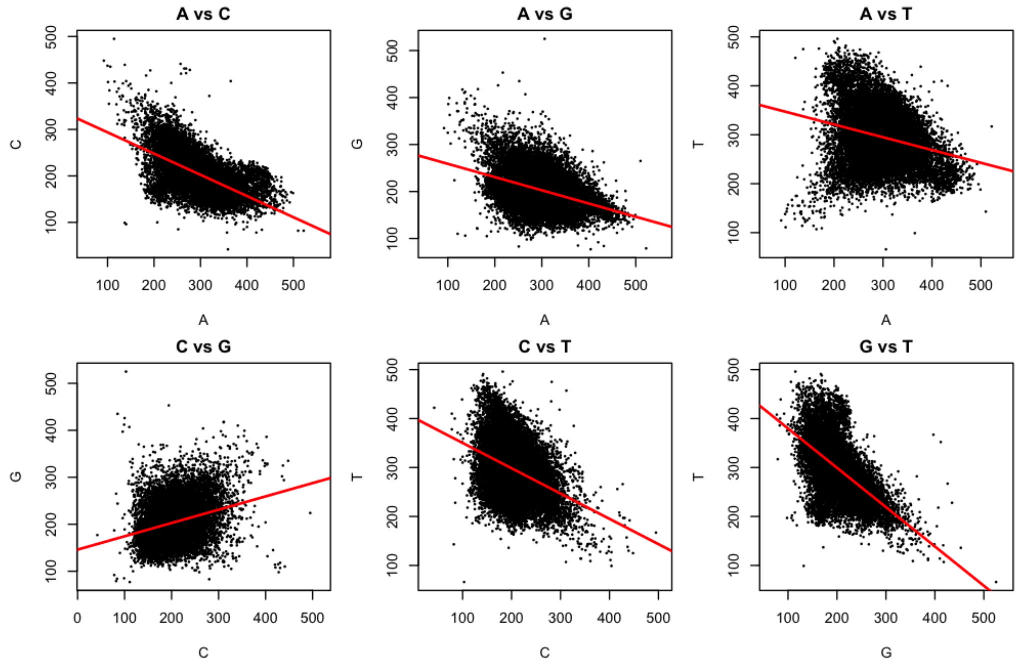

Figure 11: Lab 1 — Pairwise base-count scatter plots. Each point is one 1000-bp cell. The A–T and C–G pairs (anti-diagonal entries) show the clearest positive correlations with red regression lines. The C–G cloud clusters near the origin (lower absolute counts) but has striking upper-right outliers corresponding to the high-GC region at 53–55.5 Mb.

3.2 Lab 2 — Coverage computation and chromosome-wide distribution

We implemented and benchmarked two coverage algorithms (Figure 1). Full-chromosome binned coverage (10 kb bins, Figure 4) revealed:

- **Centromere gap** at ∼120–130 Mb: coverage collapses to near zero (low-complexity, poorly-mappable repetitive sequence).
- **Outlier spikes** at ∼130–145 Mb: extreme coverage likely caused by reads mapping to duplicated segments.
- **Elevated coverage** at 0–30 Mb: also suggestive of copy-number variation or mappability artefacts.

A Normal fit to the binned coverage distribution was poor: heavy left tail, missing right tail, too flat in the middle.

3.3 Lab 3 — Testing the uniform Poisson model

Region 50–60 Mb, bins of 5000 bp. Results:

- Base-level mean coverage $\lambda \approx$ **0.036** reads/base.
- Observed vs. Poisson mismatch (Figure 2): distribution is less left-skewed and more right-skewed than Poisson.
- Binned **VMR** $\approx$ **14.7** (Poisson predicts 1); IQR $\approx 5\times$ the Poisson expectation. Decisive overdispersion.

- GC–coverage correlation $\mathbf{r} \approx \mathbf{0.84}$ over 50–100 Mb (Figure 3), consistent with the Dohm et al. paper and our professor's work. This explains the Poisson failure: coverage is driven by GC, not a uniform process.

3.4 Lab 4 — Spline regression for $f(\mathbf{gc})$

Region 50–100 Mb, 5000-bp bins. Four B-spline candidates varying knot count (1–3) and degree (2–3); all performed similarly (Figure 5). Key findings:

- R^2 capped at $\approx \mathbf{0.75}$; knot count mattered more than degree.
- Residual-SD ratio (high-GC quartile / low-GC quartile) $\approx \mathbf{2.25} \Rightarrow$ **heteroscedasticity**: error variance grows with coverage.
- Learning curve (Figure 6): RMSE drops sharply and plateaus after ∼500 training bins — the GC trend is learnable from little data.

3.5 Lab 5 — The GC relationship in depth (AI-first)

(a) G vs. C vs. G+C: `~G+C` beats `~G` or `~C` alone by $\approx \mathbf{0.17}$ in R^2. G and C are nearly interchangeable globally (Chargaff), but diverge in gene-rich, high-coverage regions where C carries independent signal — visible in the GC-skew plot (Figure 8).

(b) Spatial consistency: The spatial R^2 profile (Figure 9) shows gene-rich regions reaching $R^2 \approx 0.7$, edges intermediate, and the centromere dropping to ≈ 0.3. The relationship is *not* spatially uniform, so per-region fitting is justified.

(c) Bin size: R^2 and Spearman rise with bin size in all regions (Figure 7); statistical elbow near 5 kb, also defensible biologically. Centromere paradox: Spearman ≈ 0.95 despite $R^2 \approx 0$ — rank order is preserved but there is no real signal; R^2 is the honest metric.

(d) Poisson GLM: Poisson $\approx$ quasi-Poisson fitted curves; residual dispersion $\phi \approx \mathbf{14}$ after GC is modelled (below raw VMR ≈ 14.7 — GC explains part of the overdispersion, not all).

3.6 Lab 6 — Copy-number estimation from one sample

Bins of 2500 bp; spline fit on 50–100 Mb. Applying the plug-in estimator to 1000 consecutive bins in the gene-rich region (61.25–63.75 Mb), Figure 12:

- Histogram centered at 1 — GC bias successfully removed.
- **MAE** $\approx \mathbf{0.15}$ from the baseline: each bin deviates 15% on average, reflecting residual noise.

Full-chromosome variance decomposition (excluding centromere, telomeres, zero-read bins):

$$\text{Poisson: } \approx 16\%, \qquad \epsilon \text{ (technical): } \approx 84\%,$$

with $\hat{\sigma}^2 \approx 277$ (read units) and total $\mathrm{Var}(\hat{a}_k) \approx 0.09$. **Conclusion:** most uncertainty is technical, so a better technical model — not more reads — is the path forward. We argued the multiplicative-noise model is more biologically faithful.

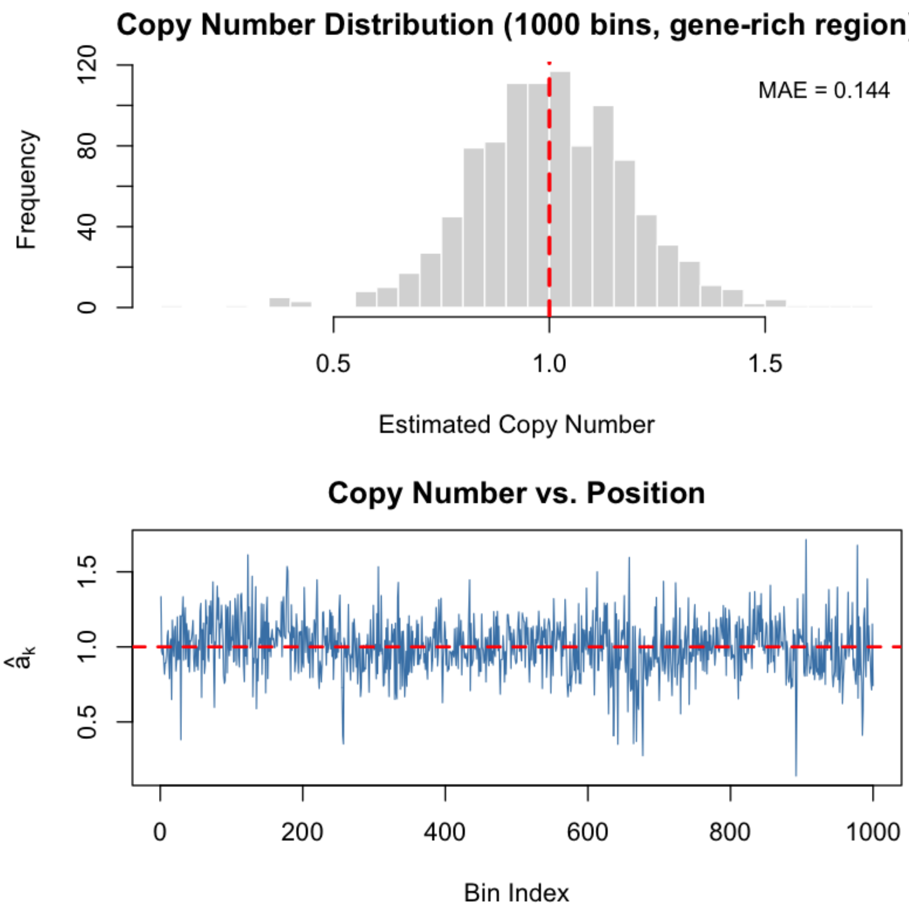

Figure 12: Lab 6 — Copy-number estimates, healthy sample (61.25–63.75 Mb). *Top:* Marginal distribution of $\hat{a}_k$ across 1000 bins. The histogram is centered at 1 (dashed red line), confirming successful GC-bias removal. MAE $\approx$ 0.15. *Bottom:* Spatial plot of $\hat{a}_k$ vs. bin index. Estimates fluctuate around 1 without systematic trends, consistent with normal copy number in this region.

3.7 Lab 7 — Tumor vs. healthy and aggregative bin-size selection

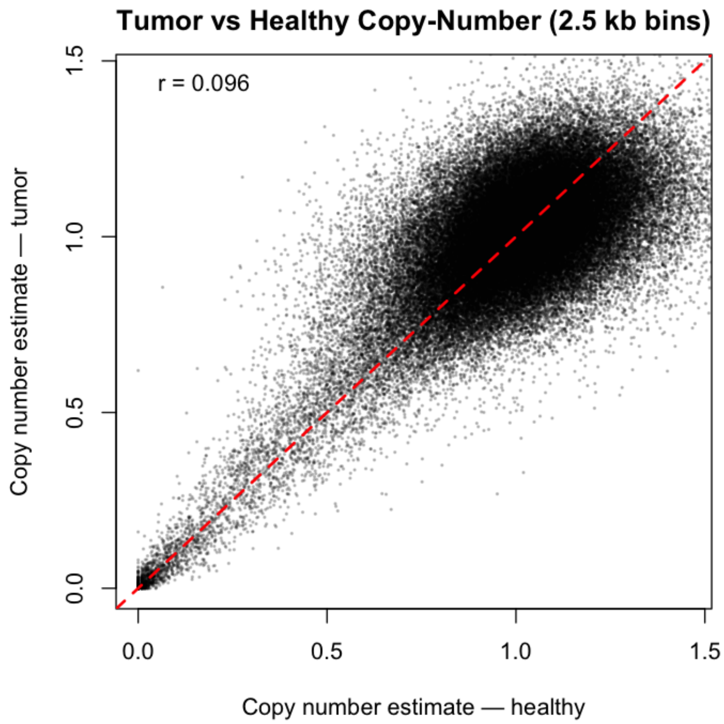

Figure 13: Lab 7 Q1 — Tumor vs. healthy copy-number scatter (2.5 kb bins). Each point is one genomic bin. The correlation is very weak ($r = 0.096$), reflecting tumor-specific CN changes plus substantial bin-level noise at 2.5 kb resolution. The red diagonal is the line of equality. Both samples are normalized to median = 1.

Q1 — Paired comparison. Healthy and tumor samples (`lib2`), 2500-bp bins, two chromosome arms (A: 5–119.99 Mb, B: 130–242 Mb; lengths chosen as multiples of 10 kb so all bin sizes align). Separate GC splines per sample (R^2: healthy **0.629**, tumor **0.484**; the tumor curve is less stable at GC extremes, reflecting greater heterogeneity). After per-sample median-normalized correction, the bin-level tumor-vs-healthy CN correlation is $r \approx 0.10$ (Figure 13) — tumor-specific CN changes plus 2.5-kb bin noise.

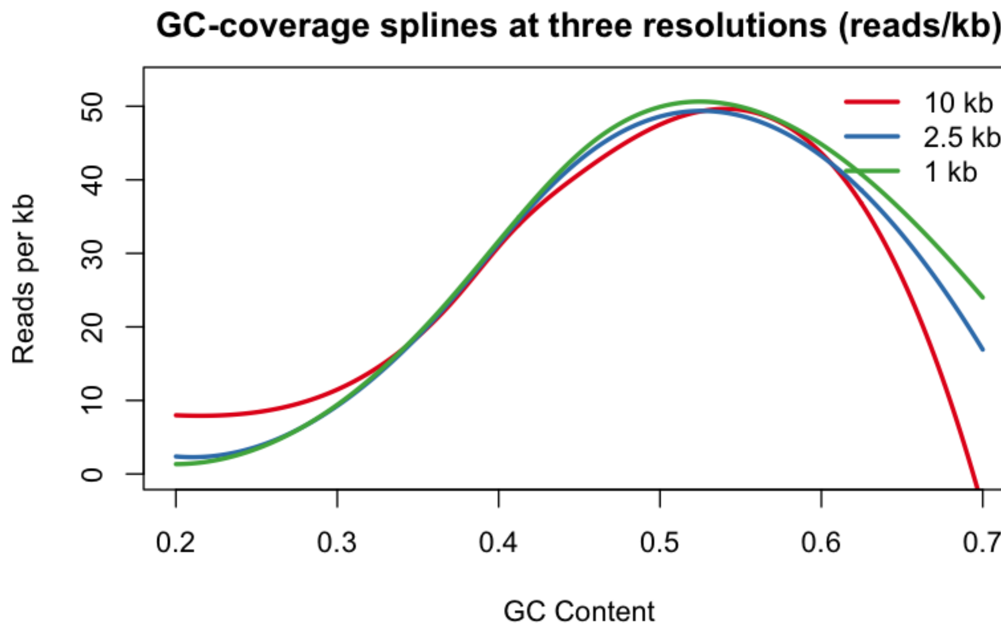

Figure 14: Lab 7 Q2 — GC–coverage splines at three resolutions (reads/kb). Red: 10 kb model; blue: 2.5 kb; green: 1 kb. All three follow the same qualitative unimodal shape. Differences are largest in the low- and high-GC tails (extrapolation), where data are sparse. Expressing predictions in reads/kb removes the trivial bin-size scaling and reveals the residual shape differences.

Q2 — Aggregative model. GC splines fit at three resolutions (10k / 2.5k / 1k), with held-out 10-kb test cells predicted either directly or by summing 4/10 sub-cell predictions (Figure 14). Model selection followed the lab-4 candidate set (4 knot/degree variants), 80/20 fit/val split, `set.seed(22)`. **1 kb gives the best held-out test loss**: smaller bins retain local GC structure that lets the spline track the true coverage profile more accurately, and the summing step aggregates those fine predictions back to the 10-kb scale without discarding that information. This is the bias–variance tradeoff in action: larger bins cut per-bin noise but blur the local GC signal; the aggregative approach captures the best of both.

3.8 Lab 9 — Anomaly detection: CNV calling on the two-sample data

Lab 9 brought together all the earlier machinery to ask: *which genomic bins show a statistically significant tumor-specific copy-number change?* The workflow has four stages.

Test statistic (Part A). We work with the two-sample GC-corrected ratio $\hat{a}_k = \hat{a}_k^{\text{tumor}}/\hat{a}_k^{\text{healthy}}$, where each sample is corrected by its own GC spline and median-normalized (the lab-8 method). Dividing cancels shared GC bias and library-specific technical effects, leaving a cleaner CN signal. The statistic is

$$S1_k = \log_2(\hat{a}_k).$$

Under H_0 (no tumor-specific CN change, ratio = 1) this is centered at 0; gains are positive, losses negative; the $\log_2$ scale is natural (value 1 = doubling, -1 = halving). We also evaluated robust window aggregates (trimmed mean S2, Huber M-estimator S3) but found them unnecessary once the negative control filter is applied (see below).

Negative controls and the empirical null (Part B). We need a direct sample from the distribution of $S1$ when H_0 is true. We apply a **local stability filter**: a bin is kept as a negative control only if its 11-bin neighbourhood has both a median near zero ($|\tilde{T}| \leq \log_2 1.10$) and low scatter (SD $\leq \log_2 1.15$). Filtering on the two-sample *ratio* (not the healthy sample alone) is correct: H_0 is "no tumor-specific change", so germline CNVs shared by both samples correctly pass the filter. About 37% of non-zero bins pass (∼31,000 controls), giving an empirical null resolving p-values down to $\approx 3 \times 10^{-5}$.

Calibration was verified by a split-half check (interleaved 100-bin blocks so null and test bins never share a window). S1 tracked the uniform diagonal almost perfectly (KS $D \approx 0.009$; Figure 15). S2/S3 were slightly conservative (over-compression caused by redundant aggregation), costing power with no compensating gain, so S1 was carried forward.

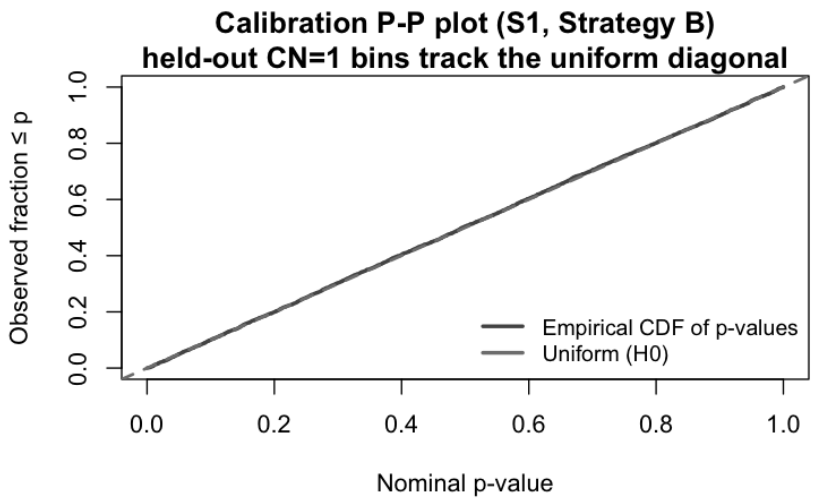

Figure 15: Lab 9 — Calibration P-P plot. Empirical CDF of held-out CN=1 bin p-values vs. the uniform diagonal. S1 tracks the diagonal closely, confirming the empirical null is well-calibrated: the realised false-positive rate matches the nominal α at every level.

P-values and multiple testing (Parts C–D). For each testable bin the two-sided empirical p-value is

$$p_k = \frac{\left|\{j \in \text{ctrl} : |S1_j| \geq |S1_k|\}\right|}{|\text{ctrl}|}.$$

Comparing empirical vs. parametric (Normal) p-values: the two methods agree on the ∼2,340 most extreme bins; the empirical test keeps 573 additional marginal bins that the Normal model drops, while the parametric assigns single-bin p-values as small as 10^{-133} — a precision the 31,000 control observations cannot support.

Multiple testing. With ∼85,000 bins tested, Bonferroni would collapse the per-test threshold to $\approx 10^{-7}$ and eliminate almost all true events. We use the **Benjamini–Hochberg (BH) procedure** instead, which controls the **False Discovery Rate** (FDR): the expected proportion of false positives *among the significant calls*, not the probability of any single false call. At FDR target 0.01, adjacent significant bins are grouped by run-length encoding (`rle`) into continuous segments.

Results (Parts D–F). Figure 16 shows the genome-wide result. After BH correction at FDR < 0.01:

- **1,600 gain bins** and **1,313 loss bins**.
- Clearest focal **deletion near 28.5 Mb** (mean ratio ≈ 0.54, roughly a two-fold loss).
- **Cluster of losses at ∼141–142 Mb** (post-centromere).
- The centromere gap (120–130 Mb) is untested, not mis-called as a loss.

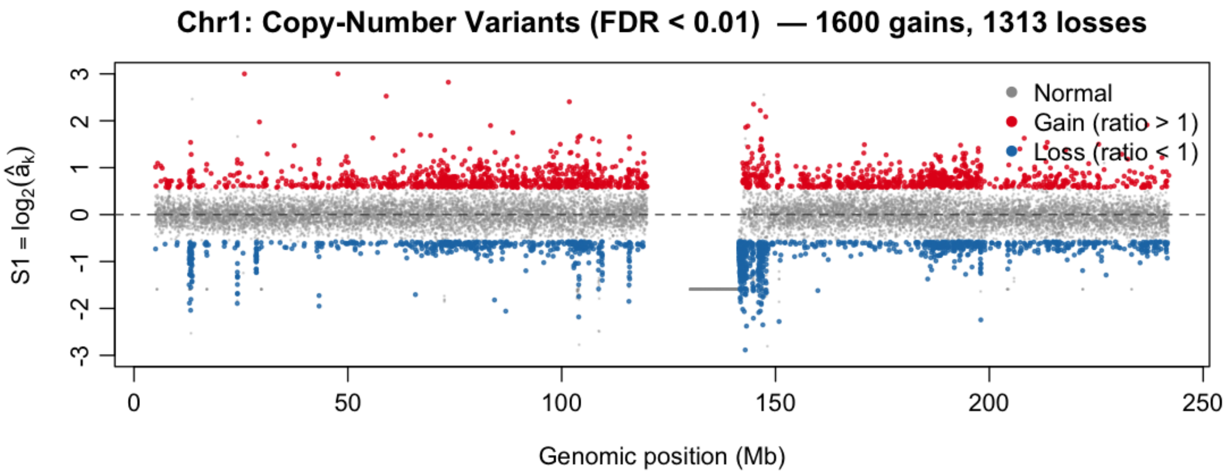

Figure 16: Lab 9 — Genome-wide copy-number variants, chr1. Each point is one 2.5 kb bin. Grey: not significant. Red: gain ($\hat{a} > 1$). Blue: loss ($\hat{a} < 1$). The centromere gap (∼120–130 Mb) is left untested (bins with zero coverage in either sample are excluded). The clearest loss cluster is near 28.5 Mb; gains are more scattered.

4 Key Numbers

Quantity	Value	Lab
Base-level mean coverage λ (50–60 Mb)	$\approx$ 0.036 reads/base	3
Binned VMR (uniform-Poisson test)	$\approx$ 14.7	3
GC–coverage Pearson correlation r	$\approx$ 0.84	3
Spline R^2 ceiling (50–100 Mb, 5 kb)	$\approx$ 0.75	4
Residual SD ratio (high/low GC quartile)	$\approx$ 2.25	4
R^2 gain from G+C vs. single base	$\approx$ 0.17	5
Quasi-Poisson dispersion ϕ (after GC)	$\approx$ 14	5
Copy-number MAE from baseline	$\approx$ 0.15	6
Variance split (Poisson / ϵ)	$\approx$ 16% / 84%	6
Total Var($\hat{a}_k$)	$\approx$ 0.09	6
GC spline R^2 (healthy / tumor)	0.629 / 0.484	7
Tumor vs. healthy CN correlation r	$\approx$ 0.10	7
Negative control bins (Strategy B)	∼31,000 (37% of nz)	9
Calibration KS D (S1, split-half)	$\approx$ 0.009	9
Significant gain bins after BH (FDR $<$ 0.01)	1,600	9
Significant loss bins after BH (FDR $<$ 0.01)	1,313	9
Focal deletion	∼28.5 Mb, ratio $\approx$ 0.54	9
High-GC anomaly region	cells 3000–5500 (∼53–55.5 Mb)	1
Centromere gap	∼120–130 Mb	2
Statistical elbow for bin size	∼5 kb	5
Default model region	50–100 Mb	3–7
Default bin size (labs 3–5)	5000 bp	3–5
Default bin size (labs 6–7)	2500 bp	6–7

5 Methods and Tools Reference

Statistical methods covered: base-composition EDA; coverage binning; the uniform Poisson model and its three assumptions; VMR and the variance decomposition for overdispersion; linear regression (`lm`) with residual diagnostics; cubic / B-spline non-parametric regression (`bs`); knot/degree model selection; train/validation/test splits, learning curves, and the bias–variance tradeoff; Poisson and quasi-Poisson GLMs (log link) with dispersion parameter ϕ; the plug-in copy-number estimator with median normalization; bias and variance of the estimator; additive vs. multiplicative noise; two-sample comparison.

R toolkit: `data.table::fread` (fast I/O), `tabulate` (fast coverage counting), `tictoc` (benchmarking), base graphics throughout with colorblind-friendly palette (`#56B4E9`/`#E69F00`), transparent points, running medians via `runmed()`, `splines::bs`, `glm`, `kableExtra`, `shiny` (interactive GC explorer, Lab 5). Shared code in `utils.R` (coverage counting) and `lab5_utils.R` (binning, spline fitting, plotting, metrics, `fit_gc_spline()`).

Foundational reference: Y. Benjamini and T. P. Speed (2012), "Summarizing and correcting the GC content bias in high-throughput sequencing," *Nucleic Acids Research* **40**(10):e72. This paper supplies the central phenomenon (GC bias is unimodal, fragment-level, PCR-driven), the coverage-as-$a_k f(\mathrm{gc}_k)$ decomposition, and the correction-by-normalization strategy that the entire lab series implements step by step.